MIL-F-3922/66B
4 May 1976
SUPERSEDING
MIL-F-3922/66A
15 June 1972

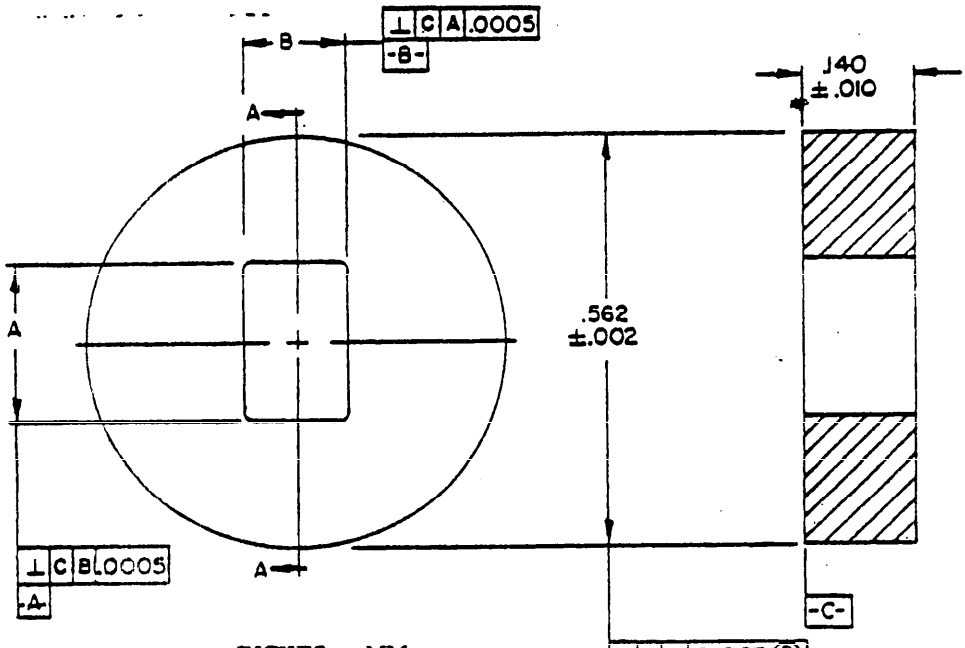
MILITARY SPECIFICATION SHEET

FLANGES, WAVEGUIDE (CONTACT)

(ROUND, 8 HOLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the flanges described herein shall consist of this document and the latest issue of Specification MIL-F-3922.



INCHES	MM
.0005	.01
.002	.05
.003	.08
.010	.25
.140	3.56
.562	14.27

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. For lettered dimensions see table I.

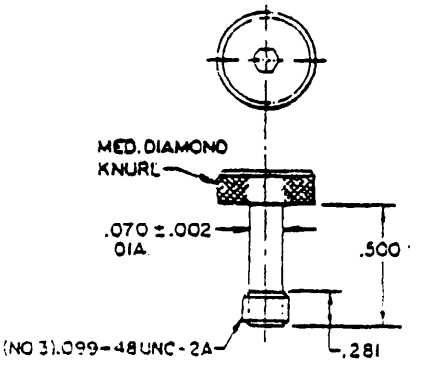
FIGURE 1. Flange blank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:08Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and dimensions.

Part No. M3922/66-			Dimensions 1/	
	Used with			
Dash No.	Waveguide M85/3-	Mating flange M3922/66-	A ±.0005 (.01)	B ±.0005 (.01)
001	020, 021, 022	001	.2050 (5.21)	.1440 (3.66)
002	017, 018, 019	002	.2310 (5.87)	.1570 (3.99)
003	047, 048, 049	003	.0930 (2.36)	.0680 (1.73)
004	044, 045, 046	004	.1070 (2.72)	.0750 (1.91)
005	050, 051, 052	005	.0850 (2.16)	.0640 (1.63)
006	041, 042, 043	006	.1220 (3.10)	.0820 (2.08)
007	023, 024, 025	007	.1830 (4.65)	.1330 (3.38)

1/ Metric equivalents are in parentheses.



INCHES	MM
.002	.05
.070	1.78
.099	2.51
.281	7.14
.500	12.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).

FIGURE 2. Socket-head cap screw.

Source: https://assist.dla.mil -- Downloaded: 202610-16T22:08Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration:
Flange - See figure 1 and table I.

Material:
Flange - Copper alloy.
Socket-head cap screw- } Corrosion resistant steel per
Hexagon nut - } QQ-S-763, 300 series.

Items supplied with flange: See table II.

AN nomenclature and engineering data: See table III.

Marking:
Part number: M3922/66- (and dash number from table I).

ENGINEERING INFORMATION:
Finished flange and waveguide assembly: See figure 3 and table IV.

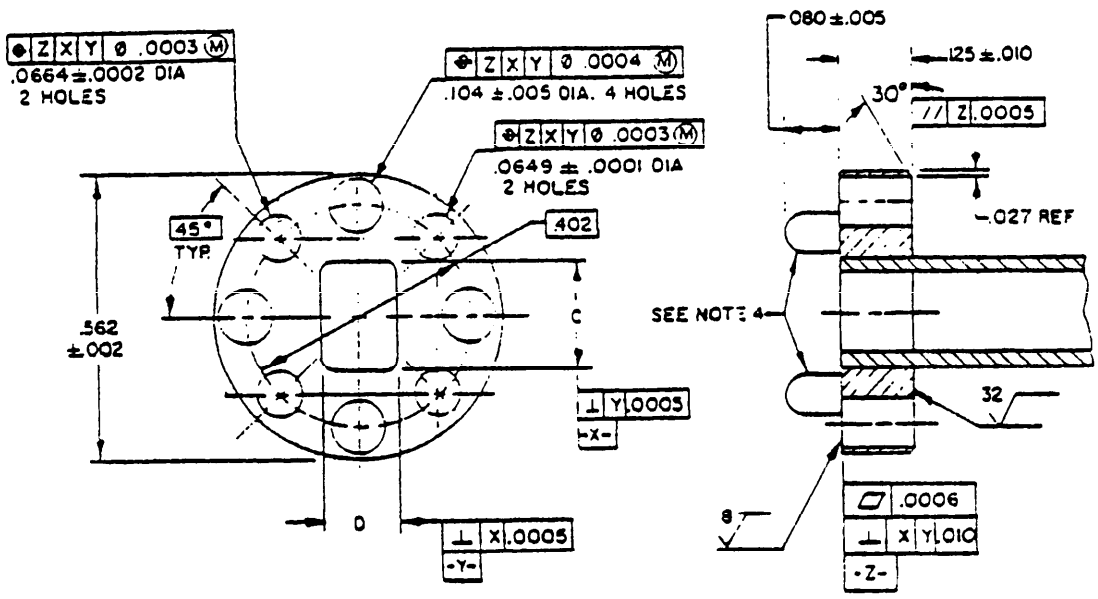
TABLE II. Items supplied with flanges.

Item	Requirement	Quantity
Socket-head cap screw	See figure 2	4
Hexagon nut	(No. 3).099-48 UNC-2B in accordance with FF-N-836, type II, style 4.	4

TABLE III. Cross-reference of part numbers/AN nomenclature and engineering data.

Part No. M3922/66-	AN nomenclature	Frequency range
		GHz
001	UG-1522/U	60.0 - 90.0
002	UG-1523/U	50.0 - 75.0
003	UG-1524/U	140.0 - 220.0
004	UG-1525/U	110.0 - 170.0
005	UG-1526/U	170.0 - 260.0
006	UG-1527/U	90.0 - 140.0
007	UG-1528/U	75.0 - 110.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:08Z
Check the source to verify that this is the current version before use.



NOTES:

1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Countersink all holes 82°, .005 $^{+.000}_{-.010}$ (.13 mm $^{+.00}_{-.25}$ mm) on front surface of flange.
4. Insert two .0655 ±.0001 (1.66 mm ±.00 mm) diameter pins of type 303 corrosion-resisting steel into .0649 (1.65 mm) holes.
5. Coupling assemblies covered by MIL-C-39004, may be used with these flanges.

TABLE IV Flange and waveguide assembly dimensions.

Part number M3922/66-	Dimensions 1/ C ±.0001 (.00)	D ±.0001 (.00)
001	.1225 (3.11)	.0615 (1.56)
002	.1490 (3.78)	.0750 (1.91)
003	.0515 (1.31)	.0260 (.66)
004	.0655 (1.66)	.0330 (.84)
005	.0435 (1.10)	.0220 (.56)
006	.0805 (2.04)	.0405 (1.03)
007	.1005 (2.55)	.0505 (1.28)

INCHES	MM	INCHES	MM
.0001	.00	.027	.69
.0002	.01	.0649	1.65
.0003	.01	.0664	1.69
.0004	.01	.080	2.03
.0005	.01	.104	2.64
.0006	.02	.125	3.18
.002	.05	.402	10.21
.005	.13	.562	14.27
.010	.25		

1/ Metric equivalents are in parentheses.

FIGURE 3. Finished flange and waveguide assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:08Z
Check the source to verify that this is the current version before use.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, SG, EL
Navy - None
Air Force - 11, 85, 80
DSA - ES

User activities:
Army - AV
Navy - AS, MC, CG, OS, SH
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0822-5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:08Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:08Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Chief of Naval Material
ATTN: MAT 0434
Washington, DC 20360

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:08Z
Check the source to verify that this is the current version before use.